

4.1 Multimodal Foundations

Joint versus coordinated spaces; alignment, fusion, and translation.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Module 2 leaves pure text. **Multimodal** systems see more than one stream: image and caption, video and audio, a PDF's pixels and its words. The hard parts are the same list as Baltrušaitis, Ahuja, and Morency: representation, translation, alignment, fusion, co-learning.

1. Why more than one stream

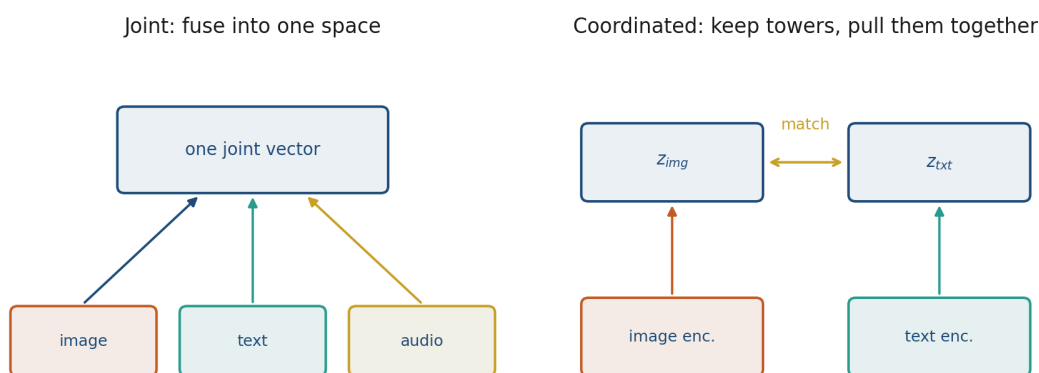
A photo of a bottle does not say "flammable." The label does. A transcript does not show who is speaking; the face does. Complementary signal, and redundant signal (lip motion plus audio). Language models in this course become **conditioned** on other modalities, or retrieve across them.

If you only train on text, you cannot answer "what is in this photo?" without an image tower or a captioner. That is why Weeks 4–5 exist.

2. Joint versus coordinated

Joint: mash the streams into **one** vector (early fusion, a concatenating MLP). Missing a modality at test time hurts.

Coordinated: keep a tower per modality, then pull the towers together with a similarity (CLIP is this). You can query with either side.



On the board: two arrows that you **add** (joint) vs two arrows that you **dot** (coordinated). CLIP never concatenates pixels with token ids during pretraining; it only scores pairs.

3. Alignment and fusion, in slogans

Alignment: which region of the image goes with which phrase. Attention is the default modern tool.

Fusion: when you combine. Early (features), late (decisions), or hybrid (cross-attention).

Translation: captioning, text-to-image, speech-to-text. Week 5's BLIP is a translation model with a matching loss.

Zero-shot / co-learning: use a rich modality to help a poor one. Week 4.3.

A detector that puts a box on "the red mug" is alignment at region level. A single cosine on the whole image vs the whole caption is alignment at instance level. Know which one your project needs.

Translation is not fusion. Captioning **maps** image $\rightarrow$ text. CLIP **scores** image $\leftrightarrow$ text. If your demo must output a sentence, you need a decoder (BLIP, Whisper, an LLM). If it must search a gallery, you need two towers and a cosine.

Co-learning: a rich labeled modality (English captions) can supervise a poorer one (a language with few labels) if they share the image. That slogan is Week 4.3's zero-shot, not a new architecture.

4. Teaching this note

30–40 minutes. Motivate two streams, then spend the block on joint vs coordinated with the 2-D cosine example. Play Yannic Kilcher CLIP **4:40–14:40** (connecting image and text; two towers as a coordinated space). Zero-shot details wait for 4.3; captioning waits for BLIP.

5. Worked example

Image vector $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Matching caption $t_+ = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$, other caption $t_- = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

$$\cos(v, t_+) = 0.8 / \sqrt{0.68} \approx 0.97, \quad \cos(v, t_-) = 0.$$

Coordinated retrieval: rank captions by cosine. Joint fusion: $\text{concat}(v, t)$ is 4-D; you need **both** at test time, and a typed query with no image has the wrong shape.

If the image is missing, a joint MLP trained on concatenations cannot score the text alone unless you trained a special "image dropout" path. A coordinated text tower still embeds the query.

6. Where students get stuck

- Concatenating then calling it CLIP.
 - Assuming a joint model can take a text-only query at test time.
 - Mixing alignment (what matches what) with fusion (when you mix features).
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7. Video

Watch [Yannic Kilcher: OpenAI CLIP, Connecting Text and Images](#).

Play **4:40–14:40**. Pause when the two encoders emit vectors in one space, and when a cosine (not a concatenating MLP) is the interaction. That is coordinated representation.

8. Practice

1. Your app must search photos with a typed query. Joint or coordinated? Why?
2. Give one example of alignment that is not “the whole image vs the whole caption.”
3. What breaks if a joint model never saw audio-only examples and you drop the image at test time?
4. Compute $\cos\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ 3 \end{bmatrix}\right)$. If you fuse by concatenation instead, what is the dimension of $\text{concat}(u, v)$?
5. Two images $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and one query $q = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$. Which image ranks first by cosine?